Avi Wadhwa

Ph.D. researcher in theoretical physics

International Centre for Theoretical Sciences, Tata Institute of Fundamental Research
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aviwadhwa.github.io | INSPIRE | ORCID | arXiv

RESEARCH INTERESTS

Post-Newtonian dynamics and worldline effective field theory; AdS/CFT and holography; scattering amplitudes, soft theorems and gauge invariance in flat and curved spacetime.

EDUCATION

Ph.D. in Physics 2021–present
International Centre for Theoretical Sciences, TIFR, Bengaluru, India
Thesis topic: *Post-Newtonian dynamics and AdS/CFT*
Advisor: R. Loganayagam

Integrated M.Sc. (Hons.) in Physics 2014–2018
Birla Institute of Technology and Science, Pilani – K.K. Birla Goa Campus, India
Thesis: *Aspects of Superconformal Representation Theory in Four Dimensions*
Thesis advisor: Abhijit Gadde, TIFR, Mumbai

PUBLICATIONS AND PREPRINTS

[1] **Post-Newtonian theory in de Sitter and anti-de Sitter spacetimes**
R. Loganayagam and **Avi Wadhwa**
arXiv:2610.20589 [hep-th] | Preprint, 2026

[2] **Subleading Chern–Simons soft factors in perturbative de Sitter**
Pratik Chattopadhyay and **Avi Wadhwa**
arXiv:2605.05130 [hep-th] | Preprint, 2026

[3] **Soft theorems in Chern–Simons matter theories**
Avi Wadhwa
Physical Review D **113**, 045017 (2026) | arXiv:2509.02542

Coauthor names are listed alphabetically, following the convention in theoretical high-energy physics.

TEACHING AND ACADEMIC ENGAGEMENT

Teaching assistant – Summer School on Gravitational Wave Astronomy 2026
ICTS–TIFR, Bengaluru. Tutorials for Jan Steinhoff's course, *Effective Field Theory Approach to Post-Newtonian Gravity*.

Journal club lectures – ICTS String Journal Club 2025
Delivered lectures on conformal field theory and the conformal bootstrap.

Teaching assistant – Introduction to General Relativity 2023
ICTS–TIFR, Bengaluru. Tutorials for the graduate course taught by Ashoke Sen.

CONFERENCE PRESENTATIONS

2026 **Strings 2026,** Shanghai, China. *Poster presentation.*

2026 **Pre-Strings School,** SIMIS, Shanghai, China. *Poster presentation.*

2025 **Indian Strings Meeting,** NISER and IIT Bhubaneswar, India. *Poster presentation.*

2025 **28th CAPRA Meeting on Radiation Reaction in General Relativity,** University of Southampton, United Kingdom. *Flash talk.*

RESEARCH VISITS

2020 **Indian Institute of Science,** Bengaluru, India. Visiting student.

2019 **Institute of Physics,** Bhubaneswar, India. Visiting student.

2018 **Tata Institute of Fundamental Research,** Mumbai, India. Visiting student for undergraduate thesis; advisor: Abhijit Gadde.

FELLOWSHIPS AND AWARDS

2021 **ICTS Graduate Research Fellowship.**

2021 **CSIR Junior Research Fellowship.**

2016 **National top 1%, National Graduate Physics Examination.** Merit Certificate.

2014 **CBSE Central Sector Scheme of Scholarship** for College and University Students.

SCHOOLS, PROGRAMS AND CONFERENCES ATTENDED

2026 **Summer School on Gravitational Wave Astronomy.** ICTS–TIFR, Bengaluru.

2026 **School for Advanced Topics in Particle Physics:** QCD, Effective Field Theories, and Nuclear Physics. ICTS–TIFR, Bengaluru.

2025 **From Asymptotic Symmetries to Flat Holography:** Theoretical Aspects and Observable Consequences. INFN Galileo Galilei Institute, Florence, Italy.

2025 **Loop-the-Loop-2:** Feynman calculus and its applications to gravity and particle physics. Online.

2025 **A Hundred Years of Quantum Mechanics.** ICTS–TIFR, Bengaluru.

2025 **Strings 2025, Annual Meeting.** Online.

2024 **Hearing beyond the Standard Model with Cosmic Sources of Gravitational Waves.** ICTS–TIFR, Bengaluru.

2024 **Quantum Information, Quantum Field Theory, and Gravity.** ICTS–TIFR, Bengaluru.

2023 **Field Theory and Turbulence.** ICTS–TIFR, Bengaluru.

2023 **Indian Strings Meeting.** IIT Bombay, India.

2022 **Non-perturbative and Numerical Approaches to Quantum Gravity, String Theory and Holography.** ICTS–TIFR, Bengaluru.

2022 **IFCAM Summer School on Mathematical Aspects of Quantum Mechanics.** IFCAM, Indian Institute of Science, Bengaluru.

2021 **Physics of the Early Universe.** ICTS–TIFR, Bengaluru.